

SYNOPTICS

SVKMs NMIMS
MUKESH PATEL SCH.OF TECH.MGMT ENGG-NM-M

Programme: B. Tech (AI)/B. Tech (AI & ML)/B. Tech (COMP)/B. Tech (CS)/B. Tech (CSE - CS)/B. TECH (CSE - DS)/B. Tech (DS)/B. Tech (EXTC)/B. Tech Intg. (CE)/B. Tech Intg. (DS)/B. Tech Intg. (IT)/B. Tech Intg. (Mech)/B. Tech (MECH)/B. Tech (MXTC)
Year: III / V/Semester IX/Semester V(Exam Year: 2025-2026)

Subject: Elements of Project Management

Date: 11 Dec 2025

Time: 02:00 pm to 05:00 pm (03:00 Hrs.)

Max. Marks: 100

FINAL EXAMINATION (Acad. Year:2025-2026)

Instructions:

1. This question paper contains 10 pages.
2. Answer to each new question to be started on a fresh page.
3. Figure in right hand side indicates full marks
4. Question No. 1 is compulsory
5. Out of remaining questions, attempt any 4 questions
6. In all 5 questions to be attempted
7. All questions carry equal marks
8. Standard Normal Cumulative Probability Table attached
9. Assume suitable data if necessary

Q1		[20]
a	Classification: 1. Based on ownership 2. Based on investment 3. Based on sector 4. Based on objectives 5. Based on nature 6. Based on time 7. Based on function 8. Based on risks 9. Based on output 10. Based on techno-economic characteristics 11. Based on financial institution's classification [Four to five parameters:04Marks]	[04]
b	• The project manager maintains maximum project control (through the line managers) over all resources, including cost and personnel. • Policies and procedures can be set up independently for each project, provided that they do not contradict company policies and procedures. • The project manager has the authority to commit company resources, provided that scheduling does not cause conflicts with other projects. • Rapid responses are possible to changes, conflict resolution, and project needs (as technology or schedule). • Because key people can be shared, the program cost is minimized. People can work on a variety of problems; that is, better people control is possible.	[04]

	• A strong technical base can be developed, and much more time can be devoted to complex problem solving. Knowledge is available for all projects on an equal basis. • Conflicts are minimal, and those requiring hierarchical referrals are more easily resolved. • There is a better balance among time, cost, and performance. [Four to five advantages: 04 Marks]	
c	• A RACI Matrix is a simple tool used to clarify roles and responsibilities during a project or organizational change. RACI stands for R – Responsible A – Accountable C – Consulted I – Informed • It enables you to specify who is responsible, who is accountable, who should be consulted and who should be informed for every task that makes up your project [Explanation with sketch/template: 04 Marks]	[04]
d	Importance of cost estimation in project management: 1. Project Feasibility Assessment ○ Helps determine whether the project is financially viable before initiation. ○ Supports go/no-go decisions based on projected costs vs. expected benefits. 2. Resource Planning ○ Enables accurate allocation of resources (manpower, materials, equipment). ○ Prevents underfunding or over-allocation, which can derail timelines. 3. Risk Management ○ Identifies potential cost overruns early. ○ Allows for contingency planning and risk mitigation strategies. 4. Stakeholder Confidence ○ Builds trust with clients, sponsors, and management through transparency. ○ Facilitates informed decision-making and approvals. 5. Performance Measurement ○ Serves as a baseline for tracking actual costs vs. estimated costs. ○ Helps in evaluating project efficiency and cost control. [Four Points: 04 Marks]	
e	-Project risk management is the process of identifying, evaluating and planning responses to events, both negative and positive that might occur throughout the project life cycle - Examples of risk in project management 1. Safety risks are common for construction projects 2. Changes in the value of local currency during a project affect the purchasing power and budgets of projects with large international components 3. Weather-dependent projects, such as road construction or coastal projects, risk delays due to exceptionally wet or windy weather [Definition: 02Marks Examples:02Marks]	[04]

Q2
a

The **Triple Constraints** in project management—**Scope, Time, and Cost**—form the foundational framework for managing any project. Adding **Quality** as the **fourth constraint** reflects the evolving expectations of modern engineering and technology projects, especially in fields like Computer Science, IT, and Data Science.

Triple Constraints:

1. **Scope**

- Defines **what** the project will deliver.
- Includes features, functionalities, and goals.
- Changes in scope often affect time and cost.

2. **Time**

- Refers to the **schedule** for completing the project.
- Includes deadlines, milestones, and delivery dates.
- Delays can increase costs or reduce quality.

3. **Cost**

- Represents the **budget** allocated.
- Includes resources, tools, salaries, and overheads.
- Budget overruns can compromise scope or quality.

- Quality as the Fourth Constraint

- **Quality** ensures that the project **meets standards, functions reliably**, and **satisfies user expectations**.
- It is influenced by all three constraints:
 - **Reduced time** may lead to rushed work and errors.
 - **Limited budget** may restrict testing or skilled resources.
 - **Scope changes** may introduce complexity affecting performance.

-Interrelationship of Constraints

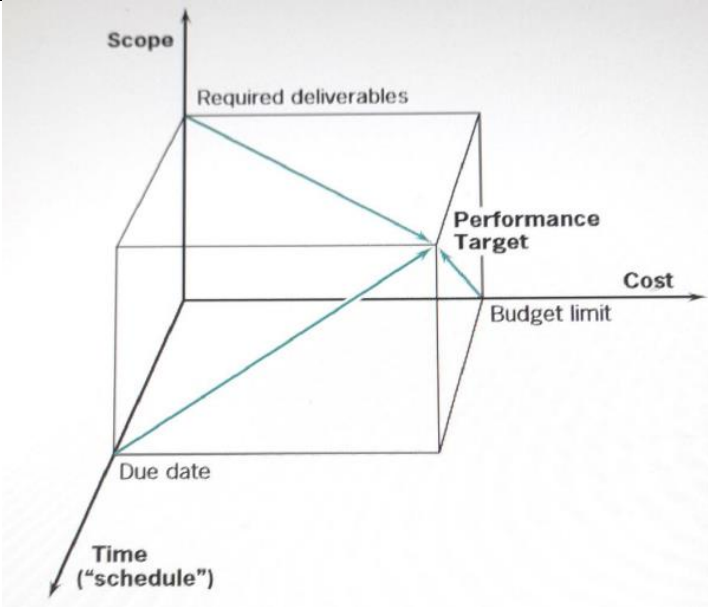
Change	Impact
Increase in scope	Requires more time and cost; may affect quality
Reduced budget	May limit resources, affecting time and quality
Tight deadlines	May lead to shortcuts, reducing quality

These constraints are **interdependent**—changing one affects the others:

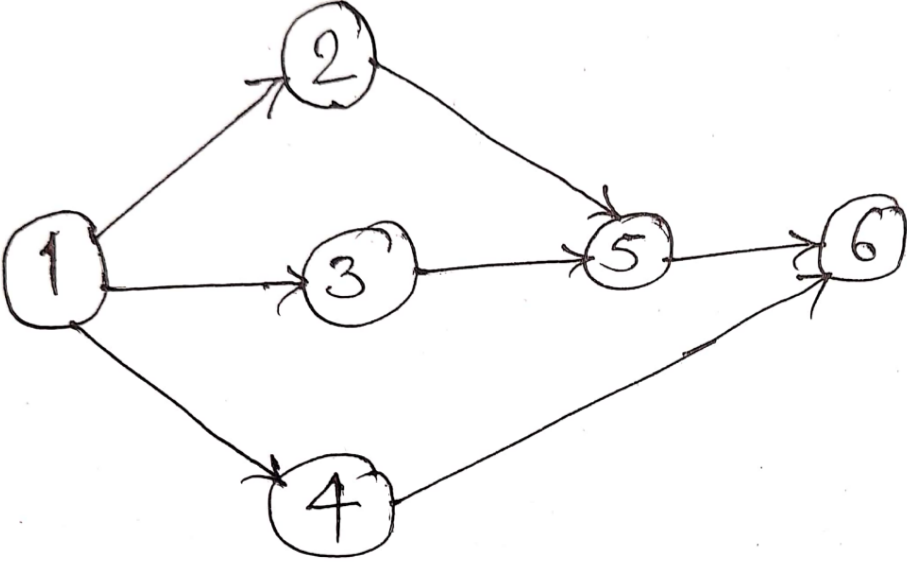
The **Triple Constraints + Quality** model helps project engineers/managers:

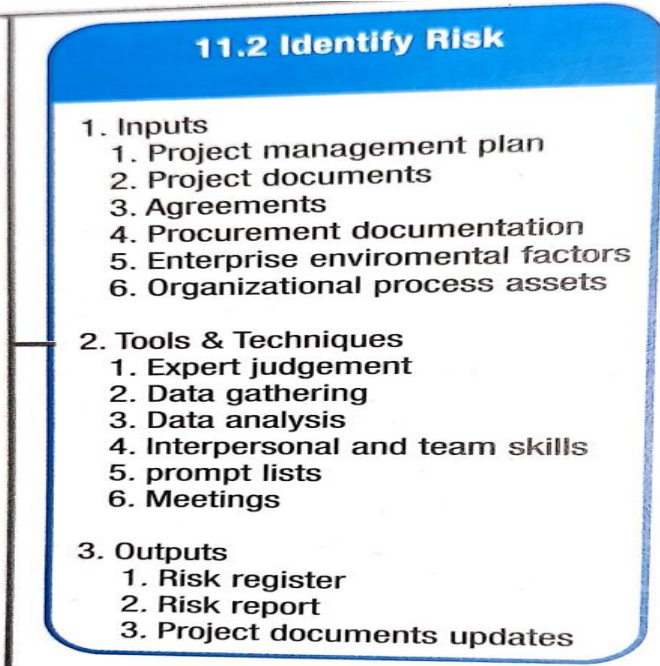
- Make informed trade-offs,
- Communicate project impacts clearly,
- Deliver successful, sustainable solutions.

[10]

	 [Explanation with example and sketch:10Marks]	
Q2 b	-Project managers are organized, goal-oriented professionals who use innovation, creativity, and collaboration to lead projects that make an impact OR -A project manager is a professional who organizes, plans, and executes projects while working within restraints like budgets and schedules - Emotional intelligence (EI) is a critical factor in the success of any project, especially when it comes to the role of the project manager. Emotional intelligence refers to the ability to: 1. Recognize, understand, and manage one's own emotions 2. Recognize, understand, and influence the emotions of others Importance of EI in Project Management- 1. <i>Improved Team Collaboration</i>	[10]

	A project manager with high EI fosters a positive team environment, encourages open communication, and resolves conflicts constructively. This leads to better collaboration and synergy among team members. <i>2. Effective Conflict Resolution</i> Projects often involve diverse stakeholders with differing opinions. EI helps the manager navigate interpersonal tensions, mediate disputes, and maintain harmony without escalating issues. <i>3. Enhanced Leadership</i> EI enables project managers to inspire and motivate their teams, adapt their leadership style to different personalities, and lead with empathy and integrity. <i>4. Better Decision-Making</i> Emotionally intelligent managers are more self-aware and less reactive. They can balance logic with empathy, leading to more thoughtful and inclusive decisions. <i>5. Stakeholder Management</i> Understanding and managing stakeholder emotions is key to building trust, negotiating effectively, and ensuring stakeholder satisfaction throughout the project lifecycle. <i>6. Stress Management</i> Projects can be high-pressure environments. EI helps managers stay calm under stress, maintain focus, and support their teams during challenging phases. Impact on Project Success- A project manager with strong emotional intelligence contributes to: • Higher team morale and productivity • Reduced turnover and burnout • Fewer misunderstandings and delays • Greater adaptability to change • Improved client and stakeholder relationships All of these factors directly influence project outcomes, including timely delivery, budget adherence, and quality of deliverables. [Definition:02Marks Explanation: 08 Marks]	
Q3 a	Explanation with example/sketch Disadvantages: -Gives only one possible schedule (earliest). -Does not show whether the project is behind schedule. -Does not demonstrate the effects of delays in any one activity on the start of another activity, thus effect on the project completion time [Explanation: 06Marks, Three to Four disadvantages:04Marks]	[10]
Q3 b	[Explanation with example and sketch:10Marks] -Decision Tree Analysis is a quantitative risk analysis technique widely used in Project Risk Management to evaluate different decision paths under uncertainty. -A decision tree is a graphical representation of decisions and their possible consequences, including chance events, costs, and payoffs. It helps project managers choose the best course of action by analyzing risks and expected outcomes -Suitability to use in quantitative risk analysis • It quantifies risks by assigning probabilities and monetary values to different outcomes. • It calculates Expected Monetary Value (EMV) for each decision path.	[10]

	• It supports objective decision-making under uncertainty. - Key Components of a Decision Tree 1. Decision Nodes (Squares) – Points where you choose among alternatives. 2. Chance Nodes (Circles) – Points where uncertain events occur with given probabilities. 3. Branches – Represent possible actions or outcomes. 4. Payoffs – Monetary values or benefits associated with each outcome. - Steps in Decision Tree Analysis 1. Define the decision problem – Identify alternatives and uncertainties. 2. Draw the tree – Start with a decision node, add chance nodes and branches. 3. Assign probabilities – For each uncertain event. 4. Assign payoffs – Estimate costs, revenues, or benefits for each outcome. 5. Calculate EMV 6. Select the best decision – The one with the highest EMV (or lowest cost). - Example in Project Risk Management with sketch	
Q4 a	1. Network diagram [02Marks] 2. Critical path: 1-3-5-6, Duration 17 weeks [02Marks] 3. Std. deviation and variance for all activities: [02Marks] 4. $Z = 1.33$, 90.82% [02 Marks] 5. Probability of not meeting due date = 25.14% [02Marks] 6. 20.84 weeks [02 Marks]  <pre> graph LR 1((1)) --> 2((2)) 1((1)) --> 3((3)) 1((1)) --> 4((4)) 2((2)) --> 5((5)) 3((3)) --> 5((5)) 4((4)) --> 6((6)) 5((5)) --> 6((6)) </pre>	[12]

Q4 b	Risk identification: -Determining which risks are likely to affect a project and documenting their characteristics -“Identify Risks” is the second process in the Project Risk Management Knowledge Area as defined by PMI’s PMBOK Guide. -It plays a crucial role in proactively recognizing potential threats and opportunities that could impact the project's objectives - Performed throughout the project lifecycle, but most intensively during the planning phase -List of Inputs to Identify Risks -List of tools & techniques -List of outputs  The diagram is a blue-bordered box with a blue header containing the text '11.2 Identify Risk'. Inside the box, there are three numbered sections: '1. Inputs' with a list of six items (Project management plan, Project documents, Agreements, Procurement documentation, Enterprise environmental factors, Organizational process assets), '2. Tools & Techniques' with a list of six items (Expert judgement, Data gathering, Data analysis, Interpersonal and team skills, prompt lists, Meetings), and '3. Outputs' with a list of three items (Risk register, Risk report, Project documents updates). [Explanation with Lists: 08Marks]	[08]
Q5 a	-The life cycle of a human-made system can be segmented into a logical series of phases and stages -System life cycle divided into four phases –called the systems development cycle 1. Conception phase (Phase A) 2. Definition phase (Phase B) 3. Execution phase (Phase C) 4. Operation phase (Phase D) Sketch and explanation for each phase [Sketch:02 Marks Explanation for each phase: 08Marks]	[10]

Q5 b	Detail procedure's steps: 1. Gathering Critical Documents 2. Identifying Key Team Members 3. Defining Level 1 Elements 4. Decomposing Elements 5. Creating WBS Dictionary 6. Creating Gantt Chart Schedule 7. WBS Charts [Detail procedure with steps supported by example:10Marks]	[10]
Q6a	-Risk register is a document that contains the results of various risk management processes and that is often displayed in a table or spreadsheet format -It is tool for documenting potential risk events and related information Importance and significance- 1. Centralized Risk Documentation The Risk Register consolidates all risk-related information in one place: • Risk descriptions • Categories • Causes and potential impacts • Probability and severity • Assigned owners • Planned responses This ensures that the project team has a clear and organized view of all risks. 2. Facilitates Risk Analysis Once risks are identified, the Risk Register supports: • Qualitative analysis (e.g., risk ranking) • Quantitative analysis (e.g., impact on cost/schedule) This helps prioritize risks based on their potential effect on project objectives. 3. Enables Proactive Risk Response Planning With risks clearly documented:	[08]

	- Teams can develop mitigation, avoidance, transfer, or acceptance strategies. - Contingency plans and fallback options can be prepared. This reduces the likelihood of surprises and improves project resilience. 4. Enhances Communication and Accountability The Risk Register: - Assigns risk owners responsible for monitoring and managing specific risks. - Serves as a communication tool among stakeholders, sponsors, and team members. - Promotes transparency and shared understanding of risk exposure. 5. Supports Monitoring and Control As the project progresses: - Risks can be updated (status changes, new risks added, resolved risks removed). - It helps track the effectiveness of risk responses. - Provides input for risk audits and lessons learned. 6. Aids in Decision-Making Project managers and stakeholders use the Risk Register to: - Make informed decisions about trade-offs and resource allocation. - Evaluate whether the project is within acceptable risk thresholds. 7. Historical Reference for Future Projects Completed Risk Registers serve as valuable organizational process assets: - They provide insights into recurring risks. - Help improve risk identification in future projects. [Explanation : 08Marks]	
Q6 b	[Four points of comparison: 04Marks]	[04]
Q6c	-Project Life Cycle Cost Estimation is a comprehensive approach in project management that involves estimating all costs associated with a project from its initiation to its closure -This method ensures that decision-makers understand the total cost of ownership (TCO) and can plan, budget, and control costs effectively throughout the project's life span -Explanation for Project life cycle cost estimation mapping/aligned with the four phases of a project: [Defining PLC cost estimation approach: 02 Marks and Mapping cost with four phases of life cycle: 06Marks]	[08]
Q7a	Project termination and closeout responsibilities- A. Planning, scheduling, and monitoring closeout activities: - Obtain and approve termination plans from involved functional managers - Prepare and coordinate termination plans and schedules - Plan for transfer of project team members and resources to other projects - Monitor completion of all contractual agreements - Monitor the disposition of any surplus materials and project equipment. B. Final closeout activities: - Close out all work orders and contracts with subcontractors and completed work - Notify all departments of project completion - Close the project office and all facilities occupied by the project organization	[08]

	• Close project books • Ensure delivery of project files and records to the responsible managers. C. Customer acceptance, obligation, and payment activities: • Ensure delivery of end-items, side-items, and customer acceptance of items • Notify the customer when all contractual obligations have been fulfilled • Ensure that all documentation related to customer acceptance as required by contract has been completed • Expedite any customer activities needed to complete the project • Transmit formal payment and collection of payments • Obtain from customer formal acknowledgment of completion of contractual obligations that release the contractor from further obligation (except warranties and guarantees). [Responsibilities in three category with brief explanation: 08Marks]	
Q7b	[Four to five points of comparison: 04Marks]	[04]
Q7c	- Evaluation for the purposes of appraising the project after it is completed and assessing the end-results is defined as summary evaluation -Summary evaluation addresses the questions “What happened?” and “What were the results?” Explanation for Project formative evaluation based on following points 1. Methods and measures 2. Site visits 3. Technology [Definition: 02 Marks, Explanation: 06 Marks]	[08]